INCH-POUND
MIL-DTL-23053/8C
17 June 1996
SUPERSEDING
MIL-I-23053/8B
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYVINYLIDENE FLUORIDE, SEMI-RIGID, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

Polymer type: The base polymer used in formulating this sleeving shall be polyvinylidene fluoride homopolymer. (see 23053/18 for co- and ter-polymers of this material.

Continuous operating temperature range: -55°C (-67°F) to +175°C (+347°F).

Color: The heat shrinkable sleeving shall be furnished transparent to translucent light tan (clear). Colors are available and shall conform to requirements of Class 1, MIL-STD-104.

Longitudinal change: ± 10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:

M23053/8 - X XX - X

- X: Color code designator (see 1.2.1)
- XX: Size identifier (see table I)
- X: Class (When no classes are specified in the specification sheet, the numeric “0” is used)
- M23053/8: Specification sheet

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Example: 0.093 inch (2.36 mm) as supplied ID sleeving shall be identified as M23053/8-003-C.

TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
M23053/8-001-C	.046 (1.17)	.023 (.58)	.010 ± .002 (.254 ± .051)
M23053/8-002-C	.063 (1.60)	.031 (.79)	.010 ± .002 (.254 ± .051)
M23053/8-003-C	.093 (2.36)	.046 (1.17)	.010 ± .002 (.254 ± .051)
M23053/8-004-C	.125 (3.18)	.062 (1.58)	.010 ± .002 (.254 ± .051)
M23053/8-005-C	.187 (4.75)	.093 (2.36)	.010 ± .002 (.254 ± .051)
M23053/8-006-C	.250 (6.35)	.125 (3.18)	.012 ± .003 (.305 ± .076)
M23053/8-007-C	.375 (9.53)	.187 (4.75)	.012 ± .003 (.305 ± .076)
M23053/8-008-C	.500 (12.70)	.250 (6.35)	.012 ± .003 (.305 ± .076)
M23053/8-009-C	.750 (19.05	.375 (9.53)	.017 ± .003 (.432 ± .076)
M23053/8-010-C	1.000 (25.40)	.500 (12.7)	.019 ± .003 (.483 ± .076)
M23053/8-011-C	1.500 (38.10)	.750 (19.1)	.020 ± .003 (.508 ± .076)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

Unrestricted shrinkage: Test method 4.6.5, 200° ± 2°C (392 ± 4°F) for 3 minutes.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 300° ± 4°C (572° ± 7.2°F)
Secant modulus, psi (MPa), minimum	120,000 (690)	4.6.12.1 ASTM D882, 2 percent strain
Restricted shrinkage	No cracks	4.6.6.1.1 175° ± 2°C (347° ± 4°F)

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Voltage withstand	Pass	4.6.6.2
Clarity stability,	Pass	4.6.16 175° ± 2°C (347° ± 4°F) for 24 hours
Concentricity	70% min.	4.6.3.3
Crystalline melt point, °C (°F), min	150 (302)	ASTM D4951 1/
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Low temperature flexibility 2/	No cracking	4.6.7.1 -55° ± 1±C (-67° ± 2°F)
Tensile strength, psi (MPa), minimum	5,000 (34.5)	4.6.13 ASTM D638, 2 inches/minute
Ultimate elongation, percent, minimum	150	4.6.13 ASTM D638, 2 inches/minute
Dielectric strength, volts/roil (Kv/mm), minimum	Sizes to .500 - 800 (31.5); Sizes over .500-600 (23.6)	4.6.2 ASTM D2671
Volume resistivity, Ohm-cm, minimum	10^{13}	4.6.2 ASTM D876
Specific gravity, max.	1.8	4.6.2 ASTM D792
Water absorption, percent, maximum	0.5	4.6.2 ASTM D570, 24 hrs at 23°C

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Heat resistance, properties after:		4.6.9 250° ± 3°C (482° ±5.4°F), 168 hours
Ultimate elongation percent, minimum	50	
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	5,000 (34.5)	
Dielectric strength, volts/roil (Kv/mm), minimum	500 (19.7)	
Flammability	3/	4.6.14 Procedure C ASTM D2671
Corrosion 4/	No corrosion	
Fungus resistance 5/	No growth	4.6.2 ASTM G21

1/ Test procedure in accordance with ASTM D4591. Melting point calculated as specified in “Melting endotherm peak” of ASTM D3222.

2/ Low temperature flexibility test specimens and mandrels: Specimens of size 0.187 inch (4.75 mm) and smaller shall be tested as whole-sections of sleeving without shrinking on conductors.supplied without shrinking on conductors. Specimens of larger sizes larger than 0.187 inch (4.75 mm) shall be 0.250 inch (6.35 mm) inch wide strip cut from as supplied sleeving, then shrunk. Mandrels shall be 10 times the specimen O.D. ±10% for tubular specimens or 10 times the specimen thickness ± 10% for strip specimens.

3/ Record whether any flaming or glowing particles or flaming drops fall from the specimen. If more than 25 percent of the indicator flag is burned away or charred (brown scorching and soot shall be ignored) after all of the five applications of flame or if flaming or glowing particles or flaming drops fall from the specimen at any time and ignite the cotton (flameless charring of the cotton shall be ignored), the specimen shall be judged to have conveyed flame and shall be considered as failing. In addition, any specimen which continues to flame or glow from one flame application longer than one minute shall not be acceptable.

4/ MIL-I-23053/8 materials are non-corrosive at operating temperatures. Performance of this requirement is at the option of the acquiring activity (see 6.2).

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5/ MIL-I-23053/8 materials do not support fungus growth. Performance of this requirement is at the option of the acquiring activity (see 6.2k).

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (95°F) for 5 years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: Heat shrinkable semi-rigid polyvinylidene fluoride sleeving is intended for wire or termination strain relief.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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